Research Seminar:
Data Analysis in Finance

VPIN and the Flash Crash, revisited on Bitcoin

A replication and extension of Andersen & Bondarenko's critique of VPIN, tested out of sample on Binance BTCUSDT, January–March 2020

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PROBLEM

An early-warning signal, and its critique

VPIN was proposed as a real-time measure of toxic order flow. Later work argued that its signal is largely mechanical.

THE CLAIM

VPIN measures order-flow toxicity

Easley, López de Prado & O'Hara (2011, 2012) report that VPIN reached extreme levels before the May 2010 Flash Crash and propose it as an early warning of liquidity crises.

THE CRITIQUE

The signal may be mechanical

Andersen & Bondarenko (2014) argue that VPIN mostly tracks trading intensity and volume heterogeneity, and that trade-direction classification adds little.

Our project

We apply the Andersen & Bondarenko diagnostics to a new asset class and a new crisis: Bitcoin during the 12–13 March 2020 crash. This is an out-of-sample test of the critique, not a replication of May 2010.

PROBLEM

The question we ask

Does tick-rule VPIN (TR-VPIN) carry incremental information about next-hour realized volatility beyond simple measures of trading activity?

- We benchmark TR-VPIN against lagged realized volatility, lagged trading volume, an uninformed U2-VPIN and a persistence forecast.
- The setting is Bitcoin spot trading on Binance, with the 12–13 March 2020 crash in the test period.
- **Hypothesis** (after Andersen & Bondarenko): most of TR-VPIN's apparent information comes from trading intensity and volume heterogeneity, not from informative trade-direction classification.

"Is VPIN measuring informed trading, or just trading?"

A metric that rises before a crash is not automatically evidence of toxic order flow: volume rises too.

DATA

Three months of Binance BTCUSDT minute bars

A calm two-month training period and a volatile test month that contains the crash.

1,361
training hours, January–February

740
out-of-sample test hours in March

48
event hours, 12–13 March crash

1,100
BTC per volume bucket, about 50 a day

Item	Value
Asset, market	BTCUSDT, Binance spot
Period	1 Jan – 31 Mar 2020, UTC
Frequency	1-minute bars + individual trades
Target	Next-hour realized volatility, %
Train / test	January–February / March

Where the data come from

Binance Vision archives (trades). Raw files stay out of Git; notebooks rebuild data/processed/ after audit checks.

DATA

Calm training months, a crash in the test month



Price, minute volume and TR-VPIN; dashed line = train/test split, shaded = 12–13 March.

What the figure shows

Price: falls from about $7,900 to below $4,000 during 12–13 March.

Volume: clusters and stays elevated for the rest of March.

TR-VPIN: usually 0.2–0.3; peaks above 0.7 during the crash, together with volume.

Source: reports/figures/baseline_tr_vpin_2020_q1.png

METHOD · PART 1

Three VPIN measures, one set of volume buckets

All three share the same fixed-volume buckets (V ≈ 1,100 BTC) and the same 50-bucket rolling window; only the trade direction differs.

1

TR-VPIN (baseline)

Each minute is signed by the tick rule: close-to-close price change; a zero change keeps the last sign.

OI = |Buy − Sell| / V
TR-VPIN = mean OI, last 50 buckets

2

U2-VPIN (benchmark)

Uninformed by construction: random trade directions; uses only how bucket volume splits across minutes (weights w_i).

$u2 = \sqrt{(\Sigma\, w_i^2)}$
U2-VPIN = mean u2, last 50 buckets

3

Exchange VPIN (extension)

Signs every individual trade with Binance's own aggressor flag (is_buyer_maker) instead of the tick rule.

Same buckets, OI formula and rolling window

Why the U2 benchmark matters

If TR-VPIN behaves like U2-VPIN, the trade-direction classification is not doing the work: the signal comes from the volume pattern alone.

METHOD · PART 2

One regression, six competing models

Every model forecasts next-hour realized volatility with OLS; only the VPIN term changes.

$$RV(t+1h) = \alpha + \beta_1 \cdot RV(t-1h) + \beta_2 \cdot \log Volume(t-1h) + \gamma \cdot VPIN(t) + \varepsilon$$

Model	Features
Controls	Lagged RV + lagged log volume
Controls_TR	Controls + TR-VPIN
Controls_U2	Controls + U2-VPIN
Controls_EX	Controls + exchange-direction VPIN
Controls_TR_EX	Controls + TR-VPIN + exchange VPIN
Persistence	Previous one-hour RV

Baseline protocol

Estimation: OLS with intercept, hourly data.

Split: fit on Jan–Feb, forecast all of March, no refit.

Metric: MSE gain vs Controls, %. Positive means VPIN helps.

CODE · PART 1

A reproducible pipeline on tested modules

Numbered notebooks orchestrate the run; the core logic lives in three small modules in src/.

1
Binance data
Vision archives or the REST API

2
Notebook 02
Download, clean and audit minutes

3
Notebooks 03–04
Buckets, tick rule, TR and U2-VPIN

4
Notebooks 05–10
Forecasts, placebo, robustness

5
reports/
17 tables and 1 figure

Module in src/	Main functions	Role
tick_rule.py	classify_minutes_tick_rule	Signs each minute's volume as buy or sell
tr_vpin.py	build_volume_buckets, calculate_tr_vpin, calculate_u2_benchmark	Fixed-volume buckets, TR-VPIN and U2-VPIN
exchange_vpin.py	build_exchange_buckets	Streams trade files in chunks; exchange VPIN

Parameters fixed in research_config.json and baseline_forecast_config.json · pandas, numpy, statsmodels, scipy, pytest

CODE · PART 2

Nine automated tests on real Binance data

pytest checks the implementation against independent recomputations, not just against itself.

Volume conserved
Total, buy and sell volume survive bucketing

Bucket boundaries
Complete buckets equal V; the residual bucket is handled

Independent recount
Buy and sell volume of the first buckets recomputed

VPIN formula
OI formula, rolling window and warm-up period

Valid ranges
VPIN within [0, 1]; availability times increase

U2 formula
U2-VPIN checked by hand on the first buckets

No look-ahead
VPIN at time t never uses future minutes

All buckets
Every complete bucket recomputed independently

Tick rule
Vectorized classifier matches a plain Python loop

tests/test_tr_vpin_real_data.py, tests/test_tr_vpin_real_extra.py

RESULTS · PART 1

TR-VPIN mirrors the uninformed benchmark

The tick-rule signal and a VPIN built with random trade directions move almost one-for-one.

0.935

correlation of TR-VPIN with U2-VPIN, March test period

0.911

same correlation in the calm January–February training period

57.5%

of March volume where the minute tick sign matches the exchange side (58.9% Jan–Feb)

The takeaway

Most of the time-series variation in TR-VPIN can be produced by the observed volume pattern alone, with trade directions drawn at random. And the minute-level tick rule agrees with the exchange's own trade direction only slightly more often than a coin flip.

Source: placebo_vpin_summary.csv, minute_tick_rule_classification.csv

RESULTS · PART 2

Do VPIN measures improve the forecast?

TR-VPIN helps a little, random directions help more, and naive persistence beats every regression.

Model	RMSE	MAE	MSE gain
TR_only	1.565	0.744	−37.29%
Controls	1.336	0.577	0.00%
Controls_TR	1.331	0.571	+0.71%
Controls_U2	1.324	0.562	+1.77%
Controls_EX	1.336	0.573	−0.02%
Controls_TR_EX	1.328	0.560	+1.14%
Persistence	1.193	0.487	+20.17%

Fixed scheme, 740 March test hours; MSE gain vs Controls.

Reading the table

TR-VPIN: +0.71% over the controls.

U2-VPIN: +1.77%, more than twice the TR gain with no order-flow information.

Persistence: +20.17%, and +29.02% without the 12–13 March event.

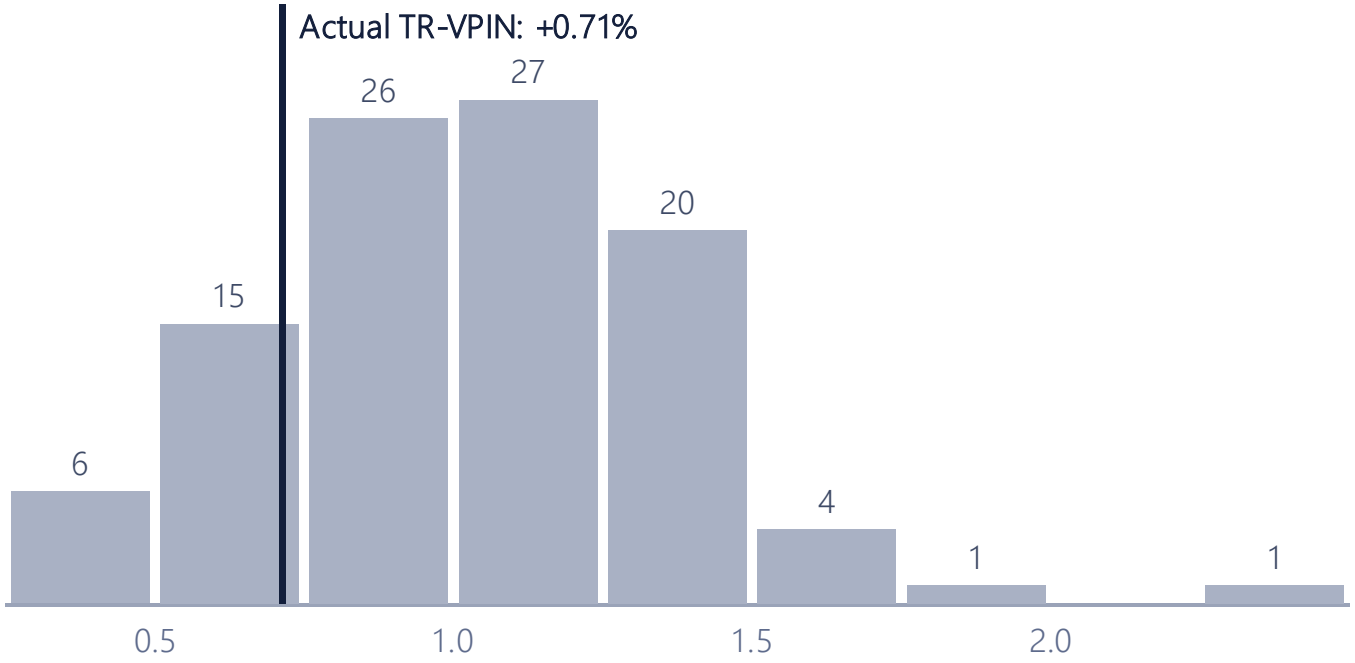
TR alone: far worse than the controls.

RESULTS · PART 3

Shuffled trade signs do as well as real ones

Trade signs are randomly permuted within each UTC day, TR-VPIN is rebuilt and the forecast refitted, 100 times.

Placebo runs (of 100) by MSE gain vs Controls, %



Bins of 0.25 pp. Source: placebo_vpin_repetitions.csv, placebo_vpin_summary.csv

80 / 100

placebo runs gain at least as much as the real TR-VPIN

1.03%

mean placebo gain; 95% range 0.42–1.70%

The 0.71% gain cannot be attributed to trade direction.

EXTENSIONS

What we added to the original framework

Six additions beyond Andersen & Bondarenko; the highlighted ones get their own slide next.

01	**New market, new crisis**	Crypto spot on Binance and the March 2020 crash, not US futures in May 2010.
02	**Exchange-direction VPIN**	Trades signed by Binance's own aggressor flag, not the tick rule (slide 13).
03	**Placebo trade signs**	100 within-day permutations of trade direction (slide 11).
04	**Re-estimation schemes**	Fixed, expanding and rolling refits, with and without the event (slide 14).
05	**Formal inference**	Block bootstrap, Clark–West and Diebold–Mariano tests, QLIKE loss (slide 15).
06	**Parameter robustness**	VPIN window, bucket size and calendar windows of 6–24 hours (slide 16).

EXTENSION · 1

Better trade directions do not rescue VPIN

Exchange-direction VPIN replaces the tick rule with Binance's record of who initiated each trade.

−0.02%

Exchange VPIN alone

MSE gain over the controls, fixed scheme. Clark–West p = 0.50: no improvement.

+1.14%

TR + exchange VPIN

Gain over the controls, but only +0.43% over TR alone (p = 0.23). The two VPINs get opposite signs: +0.70 and −1.50.

−1.48%

Not robust

Same combination under rolling re-estimation without the event: it now hurts the forecast.

Conclusion

Replacing the noisy tick rule with exchange-reported aggressor side adds no stable predictive information, so a bad classifier is not the reason VPIN fails.

Source: extended_forecast_metrics.csv, extended_regression_coefficients.csv, forecast_robustness_tests.csv

EXTENSION · 2

Gains depend on how the model is refitted

MSE gain vs Controls (%) under three estimation schemes, with and without the 48 event hours of 12–13 March.

Scheme · sample	+ TR	+ U2	+ EX	+ TR + EX	Persistence
Fixed · all March	+0.71	+1.77	−0.02	+1.14	+20.17
Expanding · all March	+1.63	+3.02	+0.06	+2.11	−9.42
Rolling · all March	+1.26	+1.95	−0.26	+1.39	−10.69
Fixed · no event	+1.42	+3.88	+1.47	+5.36	+29.02
Expanding · no event	+2.19	+4.34	+0.26	+4.65	−16.10
Rolling · no event	−0.83	−1.89	−3.84	−1.48	−14.91

Refitting helps the controls most

With expanding or rolling windows the controls overtake persistence.

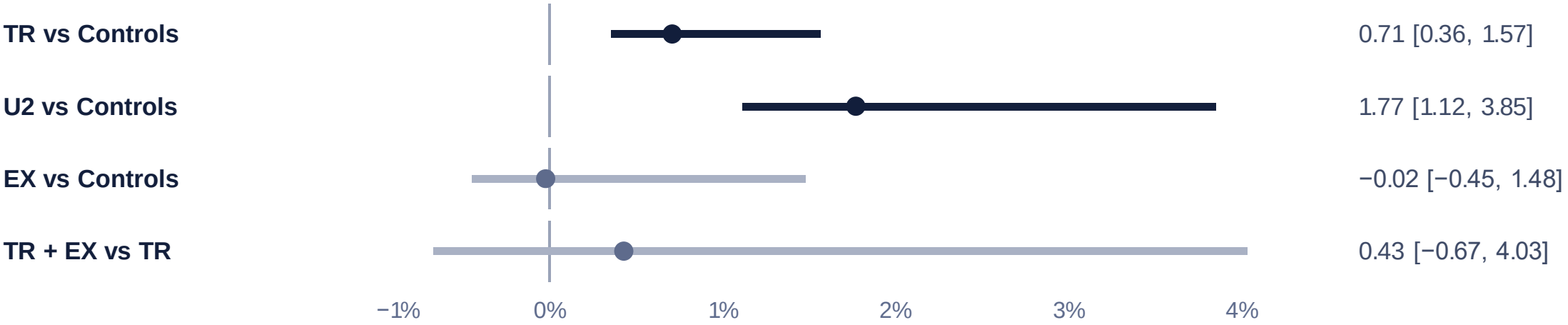
U2 beats TR in 5 of 6 rows

Under rolling refits without the event, every VPIN hurts the forecast.

EXTENSION · 3

Detectable, but economically tiny

MSE gain with 95% intervals: circular block bootstrap of March losses, 12-hour blocks, 5,000 draws, fixed scheme.



How to read it

TR's interval excludes zero, but the gain is under 1% and below the uninformed U2. Exchange VPIN intervals include zero. Results hold with 24- and 48-hour blocks.

EXTENSION · 4

U2 beats TR in all nine variants

MSE gain vs Controls when the VPIN window, bucket size or calendar window changes.

Variant	TR gain	U2 gain
Window 25 buckets	−0.16%	+0.14%
Window 50 buckets (baseline)	+0.62%	+1.72%
Window 100 buckets	+2.80%	+4.11%
Bucket 550 BTC (×0.5)	+0.03%	+0.33%
Bucket 1,100 BTC (baseline)	+0.62%	+1.72%
Bucket 2,200 BTC (×2)	+1.22%	+3.84%
Calendar window 6 h	−0.29%	−0.09%
Calendar window 12 h	−0.11%	+0.41%
Calendar window 24 h	+1.16%	+2.33%

Common robustness sample, so the baseline reads 0.62% here.

Three patterns

Uninformed always wins: random-direction U2 forecasts better than TR in every variant.

Smoother means bigger gains: longer windows and larger buckets raise both, which points to slow volume effects.

Sign can flip: short windows give TR-VPIN a negative gain.

CRITIQUE · 1

A strong core with four weak spots

The VPIN–volume link is mechanical by construction: $E[OI] = F(Q) \approx \sqrt{(2/\pi Q)}$. The weak spots are in what the paper could not test.

1 No ground truth for trade direction

Tick-rule and bulk-volume labels are never checked against true aggressor flags: the paper tests one implementation, not toxicity itself.

Our test: exchange-direction VPIN from Binance's aggressor flag (slide 13).

2 One event, one asset

The flash-crash verdict rests on a single day of E-mini data; bucket size V and window L are never varied.

Our test: new market and crisis; 9 window and bucket variants (slide 16).

3 In-sample evidence only

Full-sample OLS with no out-of-sample test; collinear regressors make the negative VPIN coefficient fragile.

Our test: Jan–Feb train, March test; DM and bootstrap (slides 14–15).

4 Target mismatch

VPIN was proposed as a liquidity-stress signal, yet it is judged only on future volatility, against a 30-day VIX.

Still open: we also forecast volatility; spread and depth are next.

Weak spots refer to Andersen & Bondarenko (2014); slide numbers point to our extensions.

CRITIQUE · 2

How could these results be overfit?

Each risk, how Andersen & Bondarenko checked it, and how we checked it on Bitcoin.

Overfitting risk	Andersen & Bondarenko	Our project
Hindsight design VPIN was built around a crash already known	Reconstructed what an observer saw before 13:30 (Figs 8–10)	Fit once on Jan–Feb, tested on unseen March
Lucky trade signs Classification may help only by chance	U1/U2-VPIN benchmarks with random trade signs	100 sign shuffles: 80 match or beat real signs
Researcher choices Bar length, bins, buckets, sample period	δ = 10/60/300 s and ν = 1/1K/5K; 2008–09 excluded; 400 shifted grids	9 window, bucket and calendar variants; 3 refit schemes

Our own overfitting risk

March 2020 was picked because it contains a crash, and the extensions were designed after seeing March. Key forecasts are repeated without the 48 event hours; the clean fix is a test on an unseen crash.

CONCLUSION

The critique carries over to Bitcoin

In BTCUSDT during March 2020, VPIN's apparent information comes mostly from trading activity, not from who was buying or selling.

What we found

- TR-VPIN closely tracks the uninformed U2-VPIN (correlation 0.91–0.94).
- Its forecast gain is 0.71%, and 80 of 100 placebo runs match or beat it.
- Exchange-reported trade direction adds no stable information.
- Gains change size and sign across windows, buckets and refit schemes.
- Naive persistence wins the fixed specification by 20%.

What our design cannot show

- One asset, one exchange, one quarter.
- March chosen because it contains a known crash.
- Extensions designed after March results were seen.
- Bootstrap ignores model-estimation uncertainty.
- Minute and exchange VPIN differ in both classification and aggregation.

REFERENCES

References and code

Main paper Andersen, T. G., Bondarenko, O. (2014). VPIN and the Flash Crash. Journal of Financial Markets, 17, 1–46.

Original VPIN Easley, D., López de Prado, M., O'Hara, M. (2011). The Microstructure of the Flash Crash. Journal of Portfolio Management, 37(2), 118–128.

Easley, D., López de Prado, M., O'Hara, M. (2012). Flow Toxicity and Liquidity in a High-Frequency World. Review of Financial Studies, 25(5), 1457–1493.

Data Binance spot BTCUSDT, 1-minute klines and trades, January–March 2020 (Binance Vision, public REST API).

Tools Python 3, pandas, numpy, statsmodels, scipy, matplotlib, pytest, JupyterLab.

Code and notebooks: github.com/yanaarsenyeva/VPIN_and_the_FlashCrash

Thank you!

Yana Arseneva, Darya Sboeva, Egor Miller, Pavel Starkov, Vladislav Gabriia